

OligoTox-Coagulopathy — Sources & Provenance

Every source, the database it came from, and a resolvable link · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

Generated 2026-10-03 from `data/sources.csv`, which is itself generated by the build. Every row of the dataset carries a `source_id` that resolves to an entry below.

100 sources	8 databases	2,685 measurements traced	100 documents held locally
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1. Databases and access routes

Every document was retrieved programmatically from a public repository. No value in this dataset comes from a search-engine summary. The endpoints below are the ones actually used, not a generic citation of the resource.

Database	What it holds	Endpoint used	Sources	Rows
PMC	PubMed Central / Europe PMC	https://www.ebi.ac.uk/europepmc/webservices/rest//fullTextXML	38	897
PubMed	NCBI PubMed (abstracts)	https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=pubmed&id=&rettype=abstract	12	65
USPTO	USPTO granted patents (full text)	https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/	10	1,264
Drugs@FDA	FDA Drugs@FDA review packages	https://www.accessdata.fda.gov/drugsatfda_docs/nda//Orig1s000.pdf	10	121
EMA	European Medicines Agency EPARs	https://www.ema.europa.eu/en/medicines/human/EPAR/	15	130
DailyMed	NLM DailyMed (US prescribing information, SPL)	https://dailymed.nlm.nih.gov/dailymed/services/v2/spls/.xml	10	97
openFDA	openFDA (FAERS adverse events, SPL labels)	https://api.fda.gov/drug/event.json?search=	4	101
other	other	—	1	10

2. How to check any single number

The chain is three steps and needs nothing from us: take a row's `source_id` from `measurements.csv`; find it in the register below or in `sources.csv`; open the link, or open the copy held at `sources/documents/<document_file>` and search for the row's `verbatim_quote` at its `source_locus`. `scripts/verify_against_sources.py` automates exactly this for every numeric value in the release.

3. Link check

Every URL in this document was requested on 2026-10-03: **174 of 185 resolved**. DOIs are checked WITHOUT following the redirect, because the question is whether the DOI is registered; following it would report a publisher's bot-blocking as a broken link. 11 did not. They are listed with their status rather than presented as working. A 401/403 means the publisher refuses automated clients, not that the record is missing. In every case the copy held at `sources/documents/` is the durable route, which is why the release ships the documents rather than only citing them.

URL	Status
https://patents.google.com/patent/US10772906	503
https://patents.google.com/patent/US10858658	503
https://patents.google.com/patent/US5886165	503
https://patents.google.com/patent/US7538211	503
https://patents.google.com/patent/US8389488	503
https://patents.google.com/patent/US9029337	503
https://patents.google.com/patent/US9061044	503
https://patents.google.com/patent/US9150864	503
https://patents.google.com/patent/US9376680	503
https://patents.google.com/patent/US9670492	503
https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9376680	no response

4. Source register

Grouped by the database the document was retrieved from. **Rows** is the number of measurements in the released dataset that cite this source; **Oligos** the number of compounds it describes.

PMC — PubMed Central / Europe PMC (38 sources)

COG-S025

156 rows · 0 oligos

Seth Chhabra E, Liu M. Evaluation of the interference of fitusiran and antithrombin lowering in plasma on routine coagulation assays. *Res Pract Thromb Haemost*. 2025;9(6):102989.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12451317/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/40988732/> **resolves** · DOI: <https://doi.org/10.1016/j.rpth.2025.102989> **resolves**

Retrieved via: PubMed Central open-access full text (JATS XML flattened to text) supplied in the coag-staging fulltext corpus

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Local copy: `sources/documents/PMC12451317.txt`

COG-S074

80 rows · 32 oligos

Nagano M, Kubota K, Sakata A, Nakamura R, Yoshitomi T, Wakui K, Yoshimoto K. A neutralizable dimeric anti-thrombin aptamer with potent anticoagulant activity in mice. *Molecular Therapy Nucleic Acids* 2023;33:659-671.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10445101/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/37621412/> **resolves** · DOI: <https://doi.org/10.1016/j.omtn.2023.07.038> **resolves**

Retrieved via: Main text: PMC JATS XML already on disk. Supplementary: the PMC OA service, the PMC /bin/ path and the Europe PMC supplementaryFiles endpoint ALL failed (404 / 500 / HTML error pages - note

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Local copy: `sources/documents/PMC10445101.xml`

COG-S038

57 rows · 1 oligos

Walsh M, Bethune C, Smyth A, Tyrwhitt J, Jung SW, Yu RZ, Wang Y, Geary RS, Weitz J, Bhanot S; CS4 Investigators. Phase 2 Study of the Factor XI Antisense Inhibitor IONIS-FXIRx in Patients With ESRD. *Kidney Int Rep*. 2022;7(2):200-209.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8820988/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/35155859/> **resolves** · DOI: <https://doi.org/10.1016/j.ekir.2021.11.011> **resolves** · ClinicalTrials.gov: <https://clinicaltrials.gov/study/NCT02553889> **resolves**

Retrieved via: PMC/Europe PMC JATS fullTextXML, staged locally; flattened for reading with an ElementTree walker

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Local copy: `sources/documents/PMC8820988.xml`

COG-S008

41 rows · 10 oligos

Ali AS et al. New High-Affinity Thrombin Aptamers for Advancing Coagulation Therapy. Cells. 2023;12(18):2230.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10526462/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/37759453/> resolves · DOI: <https://doi.org/10.3390/cells12182230> resolves

Retrieved via: PMC OA full-text JATS XML already on disk; parsed with xml.etree.

Licence: CC BY 4.0 (MDPI) | Redistribution: CC_BY

Local copy: `sources/documents/PMC10526462.xml`

COG-S073

39 rows · 2 oligos

Zavyalova E, Samoylenkova N, Revishchin A, Turashev A, Gordeychuk I, Golovin A, Kopylov A, Pavlova G. The Evaluation of Pharmacodynamics and Pharmacokinetics of Anti-thrombin DNA Aptamer RA-36. Frontiers in Pharmacology 2017;8:922.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5735248/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/29311929/> resolves · DOI: <https://doi.org/10.3389/fphar.2017.00922> resolves

Retrieved via: PMC JATS XML already on disk in the staging fulltext directory; parsed with xml.etree (single-line XML, so a real parser was used rather than grep -c). A readable rendering was written to /t

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Local copy: `sources/documents/PMC5735248.xml`

COG-S039

34 rows · 0 oligos

Buller HR, Bethune C, Bhanot S, Gailani D, Monia BP, Raskob GE, Segers A, Verhamme P, Weitz JI; FXI-ASO TKA Investigators. Factor XI Antisense Oligonucleotide for Prevention of Venous Thrombosis. N Engl J Med. 2015;372(3):232-240.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4367537/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/25482425/> resolves · DOI: <https://doi.org/10.1056/NEJMoa1405760> resolves · ClinicalTrials.gov: <https://clinicaltrials.gov/study/NCT01713361> resolves

Retrieved via: NIH author manuscript (NIHMS670480) deposited in PMC, retrieved as JATS XML. The sibling stub /home/user/coag-staging/fulltext/BULLER2015_PMC4367537.xml is 0 bytes and was NOT used.

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Local copy: `sources/documents/NCBI_PMC4367537.xml`

COG-S004

32 rows · 1 oligos

Gissel M, Orfeo T, Foley JH, Butenas S. Effect of BAX499 aptamer on tissue factor pathway inhibitor function and thrombin generation in models of hemophilia. Thromb Res 2012;130(6):948-955.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3508133/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/22951415/> resolves · DOI: <https://doi.org/10.1016/j.thromres.2012.08.299> resolves

Retrieved via: PMC NIH author-manuscript JATS XML staged locally (accepted manuscript, pre-copyedit); Table 1 recovered from in .

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Local copy: `sources/documents/PMC3508133.xml`

COG-S041

31 rows · 1 oligos

Lang T, Hodel K, Kubitz E, et al. Pharmacokinetics, pharmacodynamics, and safety of fesomersen in healthy Chinese, Japanese, and Caucasian volunteers. Clin Transl Sci. 2024;17(4):e13784.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10985948/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/38563414/> resolves · DOI: <https://doi.org/10.1111/cts.13784> resolves

Retrieved via: PMC/Europe PMC JATS fullTextXML, staged locally

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Local copy: `sources/documents/PMC10985948.xml`

COG-S027

29 rows · 0 oligos

Young G, Kavakli K, Klamroth R, et al. Safety and efficacy of a fitusiran antithrombin-based dose regimen in people with hemophilia A or B: the ATLAS-OLE study. Blood. 2025;145(25):2966-2977.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12824673/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/40053895/> resolves · DOI: <https://doi.org/10.1182/blood.2024027008> resolves

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Local copy: `sources/documents/PMC12824673.txt`

COG-S028

29 rows · 0 oligos

Pipe SW, Lissitchkov T, et al. Long-term safety and efficacy of fitusiran prophylaxis, and perioperative management, in people with hemophilia A or B. *Blood Adv*. 2025;9(5).

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11914172/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/39642315/> **resolves** · DOI: <https://doi.org/10.1182/bloodadvances.2024013900> **resolves**

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Local copy: `sources/documents/PMC11914172.txt`

COG-S042

26 rows · 2 oligos

Ferrone JD, Bhattacharjee G, Revenko AS, Zanardi TA, Warren MS, Derosier FJ, Viney NJ, Pham NC, Kaeser GE, Baker BF, Beech S, Monia BP, MacLeod AR, Wang Y, Crooke RM. IONIS-PKKRx a Novel Antisense Inhibitor of Prekallikrein and Bradykinin Production. *Nucleic Acid Ther*. 2019;29(2):82-91.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6461157/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/30817230/> **resolves** · DOI: <https://doi.org/10.1089/nat.2018.0754> **resolves**

Retrieved via: PMC/Europe PMC JATS fullTextXML, staged locally. Byte-identical duplicate present at /home/user/coag-staging/fulltext/PKK_PMC6461157.xml; flattened text copies at PMC6461157.txt and PKK_PMC6

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Local copy: `sources/documents/PMC6461157.xml`

COG-S010

25 rows · 11 oligos

Yu H, et al. Aptameric hirudins as selective and reversible EXosite-ACTive site (EXACT) inhibitors. *Nat Commun*. 2024;15:3977.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11087511/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/38730234/> **resolves** · DOI: <https://doi.org/10.1038/s41467-024-48211-6> **resolves**

Retrieved via: PMC OA full-text JATS XML already on disk; parsed with xml.etree. Supplementary Table 1 (sequences) and Supplementary Figs 1-8 not retrieved.

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Local copy: `sources/documents/PMC11087511.xml`

COG-S031

24 rows · 0 oligos

McCluskey G, Maynadie H, Borgel D, et al. Fitusiran treatment modulates the ratio between alpha- and beta-antithrombin isoforms. *HemaSphere*. 2026;10(5):e70376.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC13146349/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/42100790/> **resolves** · DOI: <https://doi.org/10.1002/hem3.70376> **resolves**

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Local copy: `sources/documents/PMC13146349.txt`

COG-S072

23 rows · 4 oligos

Braendli-Baiocco A, Festag M, Dumong Erichsen K, Persson R, Mihatsch MJ, Fisker N, Funk J, Mohr S, Constien R, Ploix C, Brady K, Berrera M, Altmann B, Lenz B, Albassam M, Schmitt G, Weiser T, Schuler F, Singer T, Tessier Y. From the Cover: The Minipig is a Suitable Non-Rodent Model in the Safety Assessment of Single Stranded Oligonucleotides. *Toxicological Sciences* 2017;157(1):112-128.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5414856/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/28123102/> **resolves** · DOI: <https://doi.org/10.1093/toxsci/kfx025> **resolves**

Retrieved via: PMC Open Access JATS XML, staged locally. Byte-identical duplicate at /home/user/coag-staging/fulltext/MINIPIG_PMC5414856.xml (md5 d813fdd2e3c336b8b4ed529813cddf95); flattened text copies at

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Local copy: `sources/documents/PMC5414856.xml`

COG-S007

21 rows · 6 oligos

Soule EE, Yu H, Olson L, Naqvi I, Kumar S, Krishnaswamy S, Sullenger BA. Generation of an anticoagulant aptamer that targets factor V/Va and disrupts the FVa-membrane interaction in normal and COVID-19 patient samples. *Cell Chem Biol*. 2022 Feb 17;29(2):215-225.e5.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8808741/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/35114109/> **resolves** · DOI: <https://doi.org/10.1016/j.chembiol.2022.01.009> **resolves**

Retrieved via: PMC OA full-text JATS XML already on disk; parsed with xml.etree. Supplementary bundle /home/user/coag-staging/supp_PMC8808741.zip was checked and is NOT a zip - it is a 10 KB EMBL-EBI 'Error

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Local copy: `sources/documents/PMC8808741.xml`

COG-S023

19 rows · 2 oligos

Abourahma H, Kempaiah P, Farooqui A, Krupa E, et al. A Comparative Study of Porcine and Ovine Derived Defibrotide: Coagulation Profiles, Molecular Characteristics, and DNA Content. Clinical and Applied Thrombosis/Hemostasis. SAGE Publications. doi:10.1177/10760296261429228

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC13009829/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/41869748/> resolves · DOI: <https://doi.org/10.1177/10760296261429228> resolves

Retrieved via: PMC Open Access JATS XML supplied in staging directory; parsed with xml.etree (single-line JATS, machine-readable elements)

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Local copy: [sources/documents/PMC13009829.xml](#)

COG-S030

18 rows · 0 oligos

Young G, Sorensen B, Dargaud Y, et al. Targeting of antithrombin in hemophilia A or B with investigational siRNA therapeutic fitusiran (ALN-AT3): results of the phase 1 inhibitor cohort. J Thromb Haemost. 2021;19(6):1436-1446.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8251589/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/33587824/> resolves · DOI: <https://doi.org/10.1111/jth.15270> resolves

Retrieved via: PubMed Central open-access full text (Wiley-converted JATS XML flattened to text) supplied in the coag-staging fulltext corpus

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Local copy: [sources/documents/PMC8251589.txt](#)

COG-S020

17 rows · 1 oligos

Stolte B, Nonnemacher M, Kizina K, Bolz S, Totzeck A, Thimm A, Wagner B, Deuschl C, Kleinschnitz C, Hagenacker T. Nusinersen treatment in adult patients with spinal muscular atrophy: a safety analysis of laboratory parameters. J Neurol. 2021;268(12):4667-4679.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8563549/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/33899154/> resolves · DOI: <https://doi.org/10.1007/s00415-021-10569-8> resolves

Retrieved via: PMC Open Access JATS XML, staged locally by the triage step

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Local copy: [sources/documents/PMC8563549.xml](#)

COG-S003

16 rows · 1 oligos

Ay C, Pabinger I, Kovacevic KD, et al. The VWF binding aptamer rondoraptivon pegol increases platelet counts and VWF/FVIII in type 2B von Willebrand disease. Blood Adv 2022;6(18):5467-5476.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9631691/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/35772170/> resolves · DOI: <https://doi.org/10.1182/bloodadvances.2022007805> resolves · ClinicalTrials.gov: <https://clinicaltrials.gov/study/NCT04677803> resolves

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Local copy: [sources/documents/PMC9631691.xml](#)

COG-S018

16 rows · 1 oligos

Maliglowka M, Dec A, Buldak L, Okopien B. The Effects of Inclisiran on the Subclinical Prothrombotic and Platelet Activation Markers in Patients at High Cardiovascular Risk. J Cardiovasc Dev Dis. 2025;12(9):355.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12470796/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/41002634/> resolves · DOI: <https://doi.org/10.3390/jcdd12090355> resolves

Retrieved via: PMC Open Access full text, staged locally by the triage step as flattened text

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Local copy: [sources/documents/PMC12470796.txt](#)

COG-S019

16 rows · 1 oligos

Zhu X, Li H, Hu C, Wu M, Zhou S, Wang Y, Li W. Safety analysis of laboratory parameters in paediatric patients with spinal muscular atrophy treated with nusinersen. BMC Pediatr. 2024;24:474.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11270951/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/39054521/> resolves · DOI: <https://doi.org/10.1186/s12887-024-04955-0> resolves

Retrieved via: PMC Open Access JATS XML, staged locally by the triage step

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Local copy: [sources/documents/PMC11270951.xml](#)

COG-S059

15 rows · 14 oligos

Ohara M, Nagata T, Hara RI, Yoshida-Tanaka K, Toide N, Takagi K, Sato K, Takenaka T, Nakakariya M, Miyata K, Maeda Y, Toh K, Wada T, Yokota T. DNA/RNA heteroduplex technology with cationic oligopeptide reduces class-related adverse effects of nucleic acid drugs. *Mol Ther Nucleic Acids*. 2024;35(3):102289.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11382116/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/39252874/> resolves · DOI: <https://doi.org/10.1016/j.omtn.2024.102289> resolves

Retrieved via: Europe PMC / PMC OA JATS XML; byline, journal, volume/issue, elocation-id and DOI all read out of in this file. Byte-identical duplicate at HDO_PMC11382116.xml; flattened text

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Local copy: [sources/documents/PMC11382116.xml](#)

COG-S029

14 rows · 0 oligos

Kenet G, Nolan B, Zulfikar B, et al. Fitusiran prophylaxis in people with hemophilia A or B who switched from prior BPA/CFC prophylaxis: the ATLAS-PPX trial. *Blood*. 2024;143(21):2256-2269.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11181353/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/38452197/> resolves · DOI: <https://doi.org/10.1182/blood.2023021864> resolves

Retrieved via: PubMed Central open-access full text (JATS XML flattened to text) supplied in the coag-staging fulltext corpus

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Local copy: [sources/documents/PMC11181353.txt](#)

COG-S001

13 rows · 2 oligos

Nimjee SM, de Lange F, Pitoc GA, Sullenger BA. Rats subject to extracorporeal membrane oxygenation have improved cardiac function following anticoagulation and reversal with factor IXa aptamer-antidote oligonucleotide pair. *Cardiol Plus* 2025;10(2):87-98.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12208382/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/40599554/> resolves · DOI: <https://doi.org/10.1097/CP9.000000000000122> resolves

Retrieved via: PMC Open Access JATS XML already staged locally; parsed with xml.etree (single-line JATS, table-wrap and fig captions extracted programmatically).

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Local copy: [sources/documents/PMC12208382.xml](#)

COG-S011

13 rows · 1 oligos

Reed CR, Bonadonna D, Otto JC, McDaniel CG, Chabata CV, Kuchibhatla M, Frederiksen J, Layzer JM, Arepally GM, Sullenger BA, Tracy ET. Aptamer-based factor IXa inhibition preserves hemostasis and prevents thrombosis in a piglet model of ECMO. *Mol Ther Nucleic Acids*. 2022;27:524-534.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8728519/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/35036063/> resolves · DOI: <https://doi.org/10.1016/j.omtn.2021.12.011> resolves

Retrieved via: PMC OA full-text JATS XML already on disk; parsed with xml.etree. Document S1 (Figures S1-S5, including the porcine aPTT dose-response Fig S3) not retrieved.

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Local copy: [sources/documents/PMC8728519.xml](#)

COG-S040

13 rows · 1 oligos

Crosby JR, Marzec U, Revenko AS, Zhao C, Gao D, Matafonov A, Gailani D, MacLeod AR, Tucker EI, Gruber A, Hanson SR, Monia BP. Antithrombotic effect of antisense factor XI oligonucleotide treatment in primates. *Arterioscler Thromb Vasc Biol*. 2013;33(7):1670-1678.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3717325/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/23559626/> resolves · DOI: <https://doi.org/10.1161/ATVBAHA.113.301282> resolves

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COG-S012

11 rows · 1 oligos

Kolyadko VN, Layzer JM, Perry K, Sullenger BA, Krishnaswamy S. An RNA aptamer exploits exosite-dependent allostery to achieve specific inhibition of coagulation factor IXa. *Proc Natl Acad Sci USA*. 2024;121(29):e2401136121.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11260126/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/38985762/> resolves · DOI: <https://doi.org/10.1073/pnas.2401136121> resolves

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COG-S017

10 rows · 2 oligos

Jongejan YK, Dirven RJ, Schrader Echeverri E, de Jong AJL, Pronk ACM, Kooijman S, Rensen PCN, Dahlman JE, Eikenboom JCJ, van Vlijmen BJM. Silencing of the von Willebrand factor gene in proatherothrombotic APOE*3-Leiden.CETP transgenic mice. Res Pract Thromb Haemost. 2025;9(1):102699.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11909755/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/40093962/> resolves · DOI: <https://doi.org/10.1016/j.rpth.2025.102699> resolves

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COG-S043

10 rows · 1 oligos

Liang J, Nilsson S, Wikstrom J, Cao H, Zheng S, Xu Q, Yan X, Ueckert S, Sun Q, Guo C, Zhang H, Liang Z, Gao S, Gan LM. Inhibition of Factor XI Using RBD4059: A Novel GalNAc-siRNA With Potent and Durable Antithrombotic Effects. JACC Basic Transl Sci. 2025;10(6):786-797.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12230477/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/40562491/> resolves · DOI: <https://doi.org/10.1016/j.jacpts.2024.12.005> resolves · ClinicalTrials.gov: <https://clinicaltrials.gov/study/NCT05653037> resolves

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COG-S061

10 rows · 3 oligos

Aupy P, Echevarria L, Relizani K, Zarrouki F, Haeberli A, Komisarski M, Tensorer T, Jouvion G, Svinartchouk F, Garcia L, Goyenvalle A. Identifying and Avoiding tcDNA-ASO Sequence-Specific Toxicity for the Development of DMD Exon 51 Skipping Therapy. Mol Ther Nucleic Acids. 2020;19:371-383.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7063478/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/31881528/> resolves · DOI: <https://doi.org/10.1016/j.omtn.2019.11.020> resolves

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COG-S021

9 rows · 1 oligos

Calcaterra IL, Santoro R, Vitelli N, Cirillo F, D'Errico G, Guerrino C, Cardiero G, Di Taranto MD, Fortunato G, Iannuzzo G, Di Minno MND. Assessment of Platelet Aggregation and Thrombin Generation in Patients with Familial Chylomicronemia Syndrome Treated with Volanesorsen: A Cross-Sectional Study. Biomedicines. 2024;12(9):2017.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11428464/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/39335531/> resolves · DOI: <https://doi.org/10.3390/biomedicines12092017> resolves

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COG-S060

8 rows · 4 oligos

Relizani K, Echevarria L, Zarrouki F, Gastaldi C, Dambrune C, Aupy P, Haeberli A, Komisarski M, Tensorer T, Larcher T, Svinartchouk F, Vaillend C, Garcia L, Goyenvalle A. Palmitic acid conjugation enhances potency of tricyclo-DNA splice switching oligonucleotides. Nucleic Acids Res. 2022;50(1):17-.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8754652/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/34893881/> resolves · DOI: <https://doi.org/10.1093/nar/gkab1199> resolves

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COG-S022

6 rows · 1 oligos

Demirjian S, Ailawadi G, Polinsky M, Bitran D, Silberman S, Shernan SK, Burnier M, Hamilton M, Squiers E, Erlich S, Rothenstein D, Khan S, Chawla LS. Safety and Tolerability Study of an Intravenously Administered Small Interfering Ribonucleic Acid (siRNA) Post On-Pump Cardiothoracic Surgery in Patients at Risk of Acute Kidney Injury. *Kidney Int Rep.* 2017;2(5):836-843.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5733816/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/29270490/> **resolves** · DOI: <https://doi.org/10.1016/j.ekir.2017.03.016> **resolves** · ClinicalTrials.gov: <https://clinicaltrials.gov/study/NCT00554359> **resolves**

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COG-S002

5 rows · 1 oligos

Moreno A, Pitoc GA, Ganson NJ, et al. Anti-PEG antibodies inhibit the anticoagulant activity of PEGylated aptamers. *Cell Chem Biol* 2019;26(5):634-644.e3.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6707742/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/30827937/> **resolves** · DOI: <https://doi.org/10.1016/j.chembiol.2019.02.001> **resolves**

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COG-S044

5 rows · 1 oligos

Preclinical safety and bleeding evaluation in swine for a small interfering RNA-lipid nanoparticle that prevents excess fibrinogen synthesis. *J Thromb Haemost.* 2025. (ABSTRACT ONLY - full text not obtainable)

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12766900/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/41076269/> **resolves** · DOI: <https://doi.org/10.1016/j.jtha.2025.09.012> **resolves**

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COG-S065

2 rows · 2 oligos

Schmidt M, Hagner N, Marco A, Koenig-Merediz SA, Schroff M, Wittig B. Design and Structural Requirements of the Potent and Safe TLR-9 Agonistic Immunomodulator MGN1703. *Nucleic Acid Ther.* 2015;25(3):130-.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4440985/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/25826686/> **resolves** · DOI: <https://doi.org/10.1089/nat.2015.0533> **resolves**

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COG-S066

2 rows · 1 oligos

Peters S, Wirkert E, Kuespert S, Heydn R, Johannesen S, Friedrich A, Mailaender S, Korte S, Mecklenburg L, Aigner L, Bruun TH, Bogdahn U. Safe and Effective Cynomolgus Monkey GLP-Tox Study with Repetitive Intrathecal Application of a TGFBR2 Targeting LNA-Gapmer Antisense Oligonucleotide as Treatment Candidate for Neurodegenerative Disorders. *Pharmaceutics.* 2022;14(1):200.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8780845/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/35057094/> **resolves** · DOI: <https://doi.org/10.3390/pharmaceutics14010200> **resolves**

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COG-S071

2 rows · 1 oligos

[Author byline not present in the retrieved file and therefore not asserted.] siRNA-mediated reduction of a circulating protein in swine using lipid nanoparticles. Mol Ther Methods Clin Dev. 2024;101258.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11109470/> **resolves** · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/38779336/> **resolves** · DOI: <https://doi.org/10.1016/j.omtm.2024.101258> **resolves**

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PubMed — NCBI PubMed (abstracts) (12 sources)

COG-S057

11 rows · 1 oligos

Sheehan JP, Lan HC. Phosphorothioate oligonucleotides inhibit the intrinsic tenase complex. Blood. 1998 Sep 1;92(5):1617-25.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/9716589/> **resolves**

Retrieved via: NCBI eutils PubMed abstract. Blood full text HTTP 403 through the proxy; figures paywalled. Duplicate copies at `pubmed_abstracts_mechanism.txt` (record 1) and `SHEEHAN1998_Blood_pmid9716589_ab`

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Local copy: `sources/documents/pubmed_9716589.txt`

COG-S062

10 rows · 2 oligos

Henry SP, Novotny W, Leeds J, Auletta C, Kornbrust DJ. Inhibition of coagulation by a phosphorothioate oligonucleotide. Antisense Nucleic Acid Drug Dev. 1997 Oct;7(5):503-10.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/9361909/> **resolves** · DOI: <https://doi.org/10.1089/oli.1.1997.7.503> **resolves**

Retrieved via: NCBI eutils PubMed abstract; no PMC record, Mary Ann Liebert (1997) full text not obtained. Duplicates at `HENRY1997_ANADD_pmid9361909_abstract.txt` and `pubmed_abstracts_mechanism.txt` (record

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Local copy: `sources/documents/pubmed_9361909.txt`

COG-S058

9 rows · 0 oligos

Sheehan JP, Phan TM. Phosphorothioate oligonucleotides inhibit the intrinsic tenase complex by an allosteric mechanism. Biochemistry. 2001 Apr 24;40(16):4980-9.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/11305914/> **resolves** · DOI: <https://doi.org/10.1021/bi002396x> **resolves**

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COG-S005

6 rows · 2 oligos

Chan MY, Cohen MG, Dyke CK, et al. Phase 1b randomized study of antidote-controlled modulation of factor IXa activity in patients with stable coronary artery disease. Circulation 2008;117(22):2865-2874.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/18506005/> **resolves** · DOI: <https://doi.org/10.1161/CIRCULATIONAHA.107.745687> **r**
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COG-S055

6 rows · 1 oligos

Younis HS, Crosby J, Huh JI, Lee HS, Rime S, Monia B, Henry SP. Antisense inhibition of coagulation factor XI prolongs APTT without increased bleeding risk in cynomolgus monkeys. Blood. 2012 Mar 8;119(10):2401-8.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/22246038/> **resolves** · DOI: <https://doi.org/10.1182/blood-2011-10-387134> **resolves**

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COG-S063

6 rows · 1 oligos

Shaw DR, Rustagi PK, Kandimalla ER, Manning AN, Jiang Z, Agrawal S. Effects of synthetic oligonucleotides on human complement and coagulation. *Biochem Pharmacol.* 1997 Apr 25;53(8):1123-32.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/9175717/> **resolves** · DOI: [https://doi.org/10.1016/s0006-2952\(97\)00091-9](https://doi.org/10.1016/s0006-2952(97)00091-9) **resolves**

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COG-S067

5 rows · 0 oligos

Henry SP, Bolte H, Auletta C, Kornbrust DJ. Evaluation of the toxicity of ISIS 2302, a phosphorothioate oligonucleotide, in a four-week study in cynomolgus monkeys. *Toxicology.* 1997 Jun 27;120(2):145-55.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/9184201/> **resolves** · DOI: [https://doi.org/10.1016/s0300-483x\(97\)03661-5](https://doi.org/10.1016/s0300-483x(97)03661-5) **resolves**

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COG-S064

4 rows · 3 oligos

Agrawal S, Rustagi PK, Shaw DR. Novel enzymatic and immunological responses to oligonucleotides. *Toxicol Lett.* 1995 Dec;82-83:431-4.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/8597089/> **resolves** · DOI: [https://doi.org/10.1016/0378-4274\(95\)03573-7](https://doi.org/10.1016/0378-4274(95)03573-7) **resolves**

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COG-S056

3 rows · 0 oligos

Yacyshyn BR, Barish C, Goff J, Dalke D, Gaspari M, Yu R, Tami J, Dorr FA, Sewell KL. Dose ranging pharmacokinetic trial of high-dose alicaforsen (intercellular adhesion molecule-1 antisense oligodeoxynucleotide) (ISIS 2302) in active Crohn's disease. *Aliment Pharmacol Ther.* 2002 Oct;16(10):1761-70.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/12269969/> **resolves** · DOI: <https://doi.org/10.1046/j.1365-2036.2002.01341.x> **resolves**

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COG-S068

3 rows · 1 oligos

Burel SA, Han SR, Lee HS, Norris DA, Lee BS, Machemer T, Park SY, Zhou T, He G, Kim Y, MacLeod AR, Monia BP, Lio S, Kim TW, Henry SP. Preclinical evaluation of the toxicological effects of a novel constrained ethyl modified antisense compound targeting signal transducer and activator of transcription 3 in mice and cynomolgus monkeys. *Nucleic Acid Ther.* 2013 Jun;23(3):213-27.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/23692080/> **resolves** · DOI: <https://doi.org/10.1089/nat.2013.0422> **resolves**

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Local copy: `sources/documents/pubmed_23692080.txt`

COG-S069

1 rows · 0 oligos

Glover JM, Leeds JM, Mant TG, Amin D, Kisner DL, Zuckerman JE, Geary RS, Levin AA, Shanahan WR Jr. Phase I safety and pharmacokinetic profile of an intercellular adhesion molecule-1 antisense oligodeoxynucleotide (ISIS 2302). *J Pharmacol Exp Ther.* 1997 Sep;282(3):1173-80.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/9316823/> **resolves**

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COG-S070

1 rows · 1 oligos

Sereni D, Tubiana R, Lascoux C, Katlama C, Taulera O, Bourque A, Cohen A, Dvorchik B, Martin RR, Tournerie C, Gouyette A, Schechter PJ. Pharmacokinetics and tolerability of intravenous trectovirsen (GEM 91), an antisense phosphorothioate oligonucleotide, in HIV-positive subjects. *J Clin Pharmacol.* 1999 Jan;39(1):47-54.

PubMed record: <https://pubmed.ncbi.nlm.nih.gov/9987700/> **resolves** · DOI: <https://doi.org/10.1177/00912709922007552> **resolves**

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USPTO — USPTO granted patents (full text) (10 sources)

COG-S035

702 rows · 14 oligos

Modulation of Factor 11 expression. United States Patent US 10,772,906 B2, Ionis Pharmaceuticals, Inc.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10772906> **resolves** · Patent record: <https://patents.google.com/patent/US10772906> **HTTP 503**

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COG-S037

156 rows · 4 oligos

Methods for modulating factor 12 expression. United States Patent US 9,150,864 B2, Isis Pharmaceuticals, Inc.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9150864> **resolves** · Patent record: <https://patents.google.com/patent/US9150864> **HTTP 503**

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COG-S075

129 rows · 8 oligos

Monia BP, Freier SM, Wancewicz EV, et al. (assignee Isis Pharmaceuticals, Inc.). Modulation of transthyretin expression. United States Patent US 9,061,044 B2, granted 23 June 2015. Example 24, 'Effect of ISIS Antisense Oligonucleotides Targeting Human Transthyretin in Cynomolgus Monkeys', subsection 'Coagulation Tests', Tables 86-88.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9061044> **resolves** · Patent record: <https://patents.google.com/patent/US9061044> **HTTP 503**

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COG-S033

87 rows · 10 oligos

Modulation of prekallikrein (PKK) expression. United States Patent US 9,670,492 B2, assignee Ionis Pharmaceuticals, Inc.; application 14/472,318-series filing 28 August 2014, published/granted 06 June 2017. Examples 8, 17, 19, 20, 21 and 22, Tables 50, 52, 78, 81, 82, 83, 84 and 85; per-position chemical notations at lines 1279, 1311 and 1343.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9670492> **resolves** · Patent record: <https://patents.google.com/patent/US9670492> **HTTP 503**

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Application number is given as the filing date line only ("Filing Date: 08/

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COG-S032

69 rows · 2 oligos

Modulation of factor 7 expression. United States Patent US 9,029,337 B2, assignee Isis Pharmaceuticals, Inc.; application 12/741959 (35 U.S.C. 371 national phase of PCT/US2008/082526), filed 05 November 2008, published/granted 12 May 2015. Examples 11-18, Tables 19, 20, 22, 24, 26, 27, 29, 30 and 31.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9029337> **resolves** · Patent record: <https://patents.google.com/patent/US9029337> **HTTP 503**

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COG-S034

55 rows · 21 oligos

Serpinc1 iRNA compositions and methods of use thereof. United States Patent US 9,376,680 B2, assignee Alnylam Pharmaceuticals, Inc.; application 14/806084 (continuation of Ser. No. 13/837,129, now US 9,127,274), filed 22 July 2015, published/granted 28 June 2016. Examples 7-14, Tables 15, 18 and 19; monomer abbreviation table at lines 1840-1960.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9376680> **no response** · Patent record: <https://patents.google.com/patent/US9376680> **HTTP 503**

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Inventor names are not present in this text conversion.

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COG-S036

31 rows · 6 oligos

Compositions and methods for inhibiting gene expression of factor XII. United States Patent US 10,858,658 B2, Arrowhead Pharmaceuticals, Inc.

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COG-S014

18 rows · 6 oligos

Crooke S.T. et al. "Antidotes to antisense compounds." United States Patent 8,389,488 B2 (Isis Pharmaceuticals, Inc.), issued 2013-03-05.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8389488> resolves · Patent record: <https://patents.google.com/patent/US8389488> HTTP 503

Retrieved via: pre-staged HTML-to-text conversion of the USPTO/FreePatentsOnline grant text supplied in /home/user/coag-staging/fulltext

Licence: US Government patent document - no copyright | Redistribution: public_domain

Local copy: sources/documents/US8389488.txt

COG-S015

9 rows · 1 oligos

Archemix Corp. "Aptamer therapeutics useful in the treatment of complement-related disorders." United States Patent 7,538,211 B2, issued 2009-05-26.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7538211> resolves · Patent record: <https://patents.google.com/patent/US7538211> HTTP 503

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COG-S013

8 rows · 1 oligos

Kandimalla E.R., Manning A., Agrawal S. et al. "Mixed backbone antisense oligonucleotides containing 2'-5'-ribonucleotide- and 3'-5'-deoxyribonucleotide segments." United States Patent 5,886,165 (Hybridon, Inc.), filed 1996-09-24, published 1999-03-23.

USPTO PDF: <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/5886165> resolves · Patent record: <https://patents.google.com/patent/US5886165> HTTP 503

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Local copy: sources/documents/US5886165.txt

Drugs@FDA — FDA Drugs@FDA review packages (10 sources)

COG-S093

43 rows · 1 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 219019, QFITLIA (fitusiran) injection, for subcutaneous use. Applicant: Genzyme Corporation. Reference ID 5560526. Approved 28 March 2025.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=219019> resolves

Retrieved via: FDA Drugs@FDA public review package (PDF) converted to text; supplied in bundle z-human-fda-reviews-a

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Local copy: sources/documents/fda_qfitlia_fitusiran_219019_IntegratedR.txt

COG-S088

20 rows · 0 oligos

Craig EM. Clinical Review (Medical Review), NDA 203568, Kynamro (mipomersen sodium) injection. U.S. Food and Drug Administration, Center for Drug Evaluation and Research, 2012-2013. Reference ID: 3221077. Contains named section 7.3.5.7 'Coagulation and Platelet Issues'.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=203568> resolves

Retrieved via: Pre-staged local text conversion of FDA Drugs@FDA NDA 203568 review PDF (fda_kynamro_mipomersen_203568_MedR.pdf, 3225496 bytes, 269 pp); text read with sed/grep, page numbers verified against

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Local copy: sources/documents/fda_kynamro_mipomersen_203568_MedR.txt

COG-S095

15 rows · 1 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 218614, TRYNGOLZA (olezarsen sodium) injection. Applicant: Ionis Pharmaceuticals Inc. Reference ID 5499859. Approved 19 December 2024.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=218614> resolves

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COG-S094

12 rows · 0 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Multi-disciplinary Review and Evaluation, NDA 217779, RYTELO (imetelstat) for injection. Applicant: Geron Corporation. Reference ID 5393603. Approved 6 June 2024.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=217779> resolves

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Local copy: sources/documents/fda_rytelo_imetelstat_217779_MultidisciplineR.txt

COG-S096

11 rows · 2 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 217388, WAINUA (eplontersen) injection. Applicant: Ionis Pharmaceuticals Inc. Reference ID 5299039. Approved 21 December 2023.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=217388> resolves

Retrieved via: FDA Drugs@FDA public review package (PDF) converted to text; supplied in bundle z-human-fda-reviews-a

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Local copy: sources/documents/fda_wainua_eplontersen_217388_IntegratedR.txt

COG-S080

7 rows · 0 oligos

Mentari E. Clinical Safety Review (Medical Review), NDA 211172, Tegsedi (inotersen). U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Reference ID: 4330662.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=211172> resolves

Retrieved via: FDA Drugs@FDA public review package; PDF downloaded and converted to text with PyMuPDF

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Local copy: sources/documents/fda_tegsedi_inotersen_211172_MedR.txt

COG-S090

6 rows · 0 oligos

Medical Review, NDA 209531, Spinraza (nusinersen) injection. U.S. Food and Drug Administration, CDER, 2016. Bound volume containing the Clinical Safety Review by Evelyn Mentari MD MS (Reference ID 4027874, includes named section 8.4.6.3 'Coagulation Laboratory Measurements'), the Safety Team Leader memorandum (Reference ID 4028770) and the Clinical Review by Rainer W. Paine MD PhD (Reference ID 4026626).

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=209531> resolves

Retrieved via: Pre-staged local text conversion of the FDA Drugs@FDA NDA 209531 review PDF (298 pp); page numbers verified with PyMuPDF.

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Local copy: sources/documents/fda_spinraza_nusinersen_209531_MedR.txt

COG-S097

4 rows · 0 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 215887, QALSODY (tofersen) injection, for intrathecal use. Applicant: Biogen. Reference ID 5163540. Approved 25 April 2023.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=215887> resolves

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Local copy: sources/documents/fda_qalsody_tofersen_215887_IntegratedR.txt

COG-S089

2 rows · 0 oligos

Clinical Pharmacology and Biopharmaceutics Review(s), NDA 203568, Kynamro (mipomersen sodium). U.S. Food and Drug Administration, CDER, 2012. Reference IDs: 3224357 and 3135183. Contains the mipomersen-warfarin drug-drug interaction study MIPO2900509 with INR pharmacodynamic results (Table 10).

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=203568> resolves

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Local copy: sources/documents/fda_kynamro_mipomersen_203568_ClinPharmR.txt

COG-S081

1 rows · 0 oligos

Clinical Pharmacology Review, NDA 211172, Tegsedi (inotersen). U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Reference ID: 4282036.

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=211172> resolves

Retrieved via: FDA Drugs@FDA public review package; PDF downloaded and converted to text with PyMuPDF

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EMA — European Medicines Agency EPARs (15 sources)**COG-S006**

22 rows · 3 oligos

Multi-record PubMed abstract file covering the REG1/REG2 (pegnivacogin RB006 / anivamersen RB007) and BAX499 (ARC19499) programmes: PMIDs 21724623, 17101847, 18284597, 20660806, 18506005, 22500821, 19404534, 21503856, 22658294, 22632032, 24263002.

EMA medicine page: <https://www.ema.europa.eu/en/medicines> resolves

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Local copy: sources/documents/pubmed_REG1_pegnivacogin_BAX499_abstracts.txt

COG-S009

22 rows · 6 oligos

Yu H, Pitoc G, Zhang M, et al. An Aptamer-Based EXACT Anticoagulant as a Sustainable, Animal-Free Alternative to Unfractionated Heparin. Adv Sci. 2026;13(4):e09867.

PMC article: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12822440/> resolves · PubMed record: <https://pubmed.ncbi.nlm.nih.gov/41053535/> resolves · DOI: <https://doi.org/10.1002/adv.202509867> resolves

Retrieved via: PMC OA full-text JATS XML already on disk; parsed with xml.etree. Supporting Information (Table S1 = sequences; Figures S1-S5) not retrieved.

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Local copy: sources/documents/PMC12822440.xml

COG-S082

21 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Dawnzera (donidalorsen). 2025.

EMA medicine page: <https://www.ema.europa.eu/en/medicines> resolves

Retrieved via: EMA EPAR public web page; PDF downloaded and converted to text with PyMuPDF

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Local copy: sources/documents/dawnzera_AR.txt

COG-S076

17 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Waylivra (volanesorsen), International non-proprietary name: volanesorsen, Procedure No. EMEA/H/C/004538/0000. 28 February 2019.

EMA medicine page: <https://www.ema.europa.eu/en/medicines> resolves

Retrieved via: EMA EPAR public web page; PDF downloaded and converted to text with PyMuPDF (pdf2txt.py in the same directory); PDF retained alongside at waylivra_EPAR_assessment_report.pdf

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Local copy: sources/documents/waylivra_EPAR_assessment_report.txt

COG-S079

11 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Tegsedi (inotersen), Procedure No. EMEA/H/C/004782/0000. 2018.

EMA medicine page: <https://www.ema.europa.eu/en/medicines> resolves

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COG-S054

7 rows · 6 oligos

NEGATIVE CONTROL (delivery-platform class) - GalNAc-conjugated siRNA US Prescribing Information: GIVLAARI (givosiran), OXLUMO (lumasiran), LEQVIO (inclisiran), AMVUTTRA (vutrisiran), RIVFLOZA (nedosiran), REDEMPLO (plozasiran).

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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COG-S077

7 rows · 0 oligos

Akcea Therapeutics Ireland Ltd. WAYLIVRA (volanesorsen) EU Risk Management Plan, CTD Module 1.8.2, Version 3.0, 24 February 2025 (published as part of the Waylivra EPAR).

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

Retrieved via: EMA EPAR public web page; PDF downloaded and converted to text with PyMuPDF

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Local copy: `sources/documents/waylivra_EPAR_RMP.txt`

COG-S053

4 rows · 4 oligos

NEGATIVE CONTROL (chemistry class) - Phosphorodiamidate morpholino oligomer exon-skipping US Prescribing Information: EXONDYS 51 (eteplirsén), VYONDYS 53 (golodirsén), VILTEPSO (viltolarsén), AMONDYS 45 (casimersén). Sarepta Therapeutics / NS Pharma.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

Retrieved via: NLM DailyMed SPL XML download, four separate labels bundled under one source key

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Local copy: `sources/documents/dailymed_eteplirsén_exondys_33bff678-7829-479e-9110-b8e33a0bc0aa.xml`

COG-S083

4 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Amvuttra (vutrisiran). 2022.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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Local copy: `sources/documents/amvuttra_AR.txt`

COG-S098

4 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use. Assessment report: Qalsody (tofersen). EMA/276404/2024, Procedure EMEA/H/C/005996/0000, 22 February 2024.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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Local copy: `sources/documents/qalsody_AR.txt`

COG-S092

3 rows · 1 oligos

Committee for Medicinal Products for Human Use (CHMP). Assessment report: Spinraza (nusinersén). Procedure No. EMEA/H/C/004312/0000. EMA/289068/2017, 21 April 2017. European Medicines Agency.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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COG-S099

3 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use. Assessment report: Rytelo (imetelstat). EMA/13310/2025, Procedure EMEA/H/C/006098/0000, 12 December 2024.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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Local copy: `sources/documents/rytelo_AR.txt`

COG-S078

2 rows · 0 oligos

European Medicines Agency. Waylivra (volanesorsen) Product Information: Annex I Summary of Product Characteristics, Annex II, Annex III Labelling and Package Leaflet.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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Local copy: `sources/documents/waylivra_EPAR_product_information.txt`

COG-S091

2 rows · 1 oligos

Committee for Medicinal Products for Human Use (CHMP). Assessment report: Kynamro (mipomersen), solution for injection 189 mg. Procedure No. EMEA/H/C/002429/0000. EMA/305826/2013, London, 21 March 2013. European Medicines Agency.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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Licence: EMA assessment report (Kynamro). Previously `unresolved`. Resolved by decision, not by discovery: the statement is absent from the held text layer and | Redistribution: `cite_and_link_only`

Local copy: `sources/documents/kynamro_AR.txt`

COG-S100

1 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use. Assessment report: Tryngolza (olezarsen). EMA/277774/2025, Procedure EMEA/H/C/006265/0000, 2025.

EMA medicine page: <https://www.ema.europa.eu/en/medicines/resolves>

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DailyMed — NLM DailyMed (US prescribing information, SPL) (10 sources)

COG-S024

34 rows · 1 oligos

DEFITELIO (defibrotide sodium) injection, for intravenous use. US Prescribing Information. Jazz Pharmaceuticals, Inc. Initial U.S. Approval 2016. SPL version 8, effective 2025-07-01 (retrieved via NLM DailyMed).

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=2c3db989-d7ad-41ed-9ebf-698dcf6c24ec> resolves

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COG-S026

23 rows · 1 oligos

QFITLIA (fitusiran) injection, for subcutaneous use. US Prescribing Information. Genzyme Corporation. SPL version 6, 2025-09-18.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=6dd2f8ac-6f90-4cbf-b197-97d74964135c> resolves

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COG-S046

9 rows · 1 oligos

RYTELO (imetelstat) for injection, for intravenous use - US Prescribing Information. Geron Corporation. Medication Guide issued 6/2024.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=b0fab7ca-e578-43c5-9df6-bdaff4182257> resolves

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COG-S045

8 rows · 1 oligos

TEGSEDI (inotersen) injection, for subcutaneous use - US Prescribing Information. Akcea Therapeutics, Inc. Initial U.S. Approval 2018; label revised 1/2024.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=8513207e-b55f-417b-9473-af785146a543> resolves

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COG-S048

7 rows · 1 oligos

DAWNZERA (donidalorsen) injection, for subcutaneous use - US Prescribing Information. Ionis Pharmaceuticals, Inc.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=3ff501e0-f75f-07da-e063-6294a90a0cb7> **resolves**

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COG-S049

7 rows · 1 oligos

SPINRAZA (nusinersen) injection, for intrathecal use - US Prescribing Information. Biogen.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=dd70cd5f-b0fc-4ba4-a5ea-89a34778bd94> **resolves**

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COG-S047

5 rows · 1 oligos

TRYNGOLZA (olezarsen) injection, for subcutaneous use - US Prescribing Information. Ionis Pharmaceuticals, Inc. Patient Information revised 06/2026.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=0f51aa8e-8475-8cf9-e063-6394a90a6848> **resolves**

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COG-S052

2 rows · 1 oligos

NEGATIVE CONTROL - ONPATTRO (patisiran) injection, lipid complex, for intravenous use - US Prescribing Information.

Alnylam Pharmaceuticals.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=e87ec36f-b4b4-49d4-aea4-d4ffb09b0970> **resolves**

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COG-S050

1 rows · 1 oligos

NEGATIVE CONTROL - WAINUA (eplontersen) injection, for subcutaneous use - US Prescribing Information. AstraZeneca / Ionis.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=d7dcb847-71dd-4fff-82d0-d43a465fc096> **resolves**

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COG-S051

1 rows · 1 oligos

NEGATIVE CONTROL - QALSODY (tofersen) injection, for intrathecal use - US Prescribing Information. Biogen.

DailyMed label: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=81356b45-1cb7-4eef-88ea-e44cc18b47c5> **resolves**

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Local copy: `sources/documents/dailymed_tofersen_qalsody_81356b45-1cb7-4eef-88ea-e44cc18b47c5.xml`

openFDA — openFDA (FAERS adverse events, SPL labels) (4 sources)

COG-S085

57 rows · 13 oligos

Locally computed FAERS disproportionality table over oligonucleotide therapeutics and coagulation MedDRA preferred terms, built from openFDA drug/event API counts (FAERS grand total 20,692,690 reports at query time). Generated by `/home/user/coag-staging/human/faers/prr.py`.

openFDA API: <https://open.fda.gov/apis/drug/event/> **resolves**

Retrieved via: openFDA drug/event API via curl; per-cell counts obtained with the exact query URL printed in each row's `exact_openfda_query` column; PRR and Yates chi-square computed locally by `prp.py`

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Local copy: `sources/documents/faers_coag_signal_table.csv`

COG-S086

32 rows · 0 oligos

Individual FAERS case records (openFDA drug/event) for ten oligonucleotide-drug x coagulation-PT pairs, up to 5 records per pair. Generated by /home/user/coag-staging/human/faers/pull_cases.py.

openFDA API: <https://open.fda.gov/apis/drug/event/> **resolves**

Retrieved via: openFDA drug/event API via curl, one query per drug x MedDRA PT pair with &limit=5; the exact query URL is stored in each group's exact_query field

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Local copy: **sources/documents/faers_case_records.json**

COG-S087

7 rows · 0 oligos

Individual FAERS case records (openFDA drug/event) for MACUGEN (pegaptanib sodium) x four coagulation MedDRA preferred terms.

openFDA API: <https://open.fda.gov/apis/drug/event/> **resolves**

Retrieved via: openFDA drug/event API via curl using patient.drug.medicinalproduct:"MACUGEN"; the exact query URL is stored in each group's exact_query field

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Local copy: **sources/documents/faers_pegaptanib_cases.json**

COG-S084

5 rows · 1 oligos

Astellas Pharma US, Inc. IZERVAY (avacincaptad pegol) intravitreal solution — US Prescribing Information, SPL version 9, effective 2026-02-27. Retrieved as an openFDA drug/label API record (openFDA index last_updated 2026-09-02).

Drugs@FDA: <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=217225> **resolves**

Retrieved via: openFDA drug/label API (api.fda.gov/drug/label.json), single-record JSON response saved to disk

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Local copy: **sources/documents/label_izervay_avacincaptad.json**

other — other (1 sources)

COG-S016

10 rows · 1 oligos

Teva Pharmaceutical Industries / OncoGenex Pharmaceuticals. "Custirsen treatment with reduced toxicity." United States Patent Application Publication US 2014/0275214 A1, published 2014-09-18.

No resolvable public identifier; identified as: US 2014/0275214 A1

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